BUILD WHAT DOESN'T EXIST YET

Relian™

Verifiable Software Modernization

Don't just migrate legacy code — prove it behaves identically, with cryptographic evidence anyone can verify.

Commercial | **State · Local · Education** | **Federal**

THE PROBLEM

Modernization is a trust problem — not just a code problem

Legacy systems still run benefits, banking, courts, and records. The hard part isn't writing new code — it's proving the new code behaves exactly like the old. Traditional tools ask you to trust a black box.

8–51 yrs

age of critical federal legacy systems still in service (GAO)

COBOL

still runs core systems as the workforce that maintains it retires

~80%

of federal IT spend goes to operating & maintaining existing systems (GAO)

The question every board, examiner, and auditor asks: "how do you know the new system does the same thing?" Relian answers it with proof.

WHAT RELIAN IS

An independent transformation-attestation substrate

Relian does one thing no migration vendor can do for itself: ***independently prove that modernized code behaves identically to the original.***

1

Deterministic transformation

A reproducible, rule-based engine converts legacy code — the same input always yields the same output, with no black-box AI in the transform path.

2

Committed benchmarking

Behavioral equivalence is measured against a test suite cryptographically committed before the solution existed — so the results can't be gamed.

3

Cryptographic attestation

Every transformation yields a signed, hash-chained proof of equivalence that anyone can independently re-derive and verify.

The product is the proof — not the pipeline.

HOW IT WORKS

The source stays home — only the proof travels

Your legacy source
stays inside your perimeter

→

Deterministic transform
reproducible conversion

→

Equivalence measured
vs. the committed benchmark

→

Signed attestation
hash-chained, Ed25519

→

Portable proof
you verify it independently

What crosses the line is a portable artifact, not your code: the equivalence proof, the measured metrics, and the inventory of exactly what was — and was not — transformed. It stays verifiable even if we're not in the room.

WHY A SUBSTRATE, NOT A SAAS

You rent a SaaS. You own a substrate's proof.

	Typical SaaS migration tool	Relian — verifiable substrate
Trust model	Trust the vendor's black box	**Verify the proof yourself**
Transformation	Non-deterministic AI; hard to reproduce	**Deterministic & reproducible — no black-box AI in the path**
Your source code	Uploaded to the vendor's cloud	**Never leaves your perimeter**
If the vendor exits	Access & assurance evaporate	**The signed proof persists & stays verifiable**
Auditability	An output plus an SLA	**Non-repudiable evidence for auditors, IG & boards**
Who it can vouch for	Only its own output	**Any vendor's migration — the independent referee**

The difference isn't hosting — it's where trust comes from, and what you're left holding.

THE REFEREE MODEL

Relian can verify anyone’s migration — even one we didn’t run

Because the attestation is independent of the transform, Relian can act as the neutral referee for modernization: point it at any vendor’s output plus a committed benchmark, and it returns a signed equivalence verdict. That makes Relian an assurance layer, not just a tool.

Independent

Assurance is decoupled from the party doing the work — the referee doesn’t play for either side.

Portable

The proof travels with the code and outlives any single vendor, tool, or contract.

Universal

One assurance layer across tools and vendors — extended to each new language pair benchmark-first.

Not just a migration tool — an assurance layer for modernization.

Proof, measured — not asserted

BER 1.0000

100% behavioral equivalence

on the sealed RELIAN-BENCH held-out suite, across supported COBOL constructs.

Reproducible

Deterministic output

the same legacy input produces byte-for-byte identical modernized code, every run.

Ed25519

Signed, re-derivable ledger

a hash-chained attestation any third party can independently reconstruct and check.

Measured or withheld

No estimated numbers

every figure in a Relian report is measured or omitted — never guessed or padded.

The evidence discipline is the compliance program — the same records satisfy an auditor, an IG, and a board.

For enterprises carrying legacy risk

Banks, insurers, healthcare payers, manufacturers, and software vendors modernizing mainframe or aging code.

Mainframe & COBOL modernization

Retire legacy risk with board- and auditor-ready proof that the new system behaves identically.

M&A & technology due diligence

Attest that an acquired codebase's modernization preserved behavior — evidence for the deal file.

Regulatory & audit assurance

Hand examiners a signed equivalence record instead of a vendor's assertion.

VALUE **Keep your source and IP in-house. Replace "trust the vendor" with evidence you own.**

STATE · LOCAL · EDUCATION

For governments modernizing citizen systems

State agencies, counties and cities, school districts, and tribal governments.

Benefits & eligibility systems

Modernize unemployment, Medicaid-eligibility, or tax systems with attestation for oversight.

Courts, DMV & licensing

Modernize case, licensing, and records systems with data kept on-premise (CJIS / PII).

School-district SIS & payroll

Modernize student-information and payroll systems with FERPA-safe data handling.

VALUE **Buy fast through cooperative purchasing. US-only, on-premise data residency. Board- and audit-ready evidence.**

FEDERAL

For agencies under modernization & oversight mandates

Federal departments and agencies with legacy IT — and the systems integrators and primes that serve them.

Critical legacy-system modernization

Produce non-repudiable equivalence evidence that satisfies IG and GAO oversight.

Air-gapped / on-premise transformation

Source never leaves the enclave and never touches a generative-AI model.

Software supply-chain integrity

Reproducible transformation plus signed provenance, aligned to a zero-trust posture.

VALUE **Attestation as audit evidence. No offshore access. Evidence discipline that supports FISMA, CMMC & FedRAMP programs.**

START HERE — LOW-RISK ENTRY

Two fixed-fee assessments

Legacy Code Assessment

$8K

A signed report you can act on:

- Code inventory & transformation-coverage map
- Complexity & risk scoring
- Migration-scope recommendation

Data Discovery

$12K

A signed report on the data behind the code:

- Field-level data dictionary & record layouts
- File inventory & program-to-data lineage
- Draft relational target schema

Prove the model on your real code before committing to a migration. Public-sector buyers can purchase through the State of Montana MAPS cooperative vehicle (SPB26-0608GW) — cooperative purchasing extends to other states, local and tribal governments, and federal agencies.

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BUILD WHAT DOESN'T EXIST YET

Prove it. Don't just promise it.

Provable

signed equivalence proof

Portable

the evidence outlives us

Private

source never leaves you

Independent

a neutral referee

Start with an assessment.

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